Week 1 question sheet

Problem sheet

Instructions

Attempt every question before consulting the hints or solutions. Show the conditions and intermediate steps needed to justify your answer. The target time is 255 minutes and the total is 100 marks. A basic scientific calculator is permitted only for `W01-Q03-PB`. Throughout this sheet, `log` denotes the natural logarithm.

Question 1 — sets and functions

Part A — `W01-Q01-PA` (5 marks; 10 minutes) Let

$$\Omega = \{1, 2, \dots, 12\}, \qquad A = \{x \in \Omega : x \text{ is even}\}, \qquad B = \{x \in \Omega : x \text{ divides } 12\}.$$

List $A \cap B$, $A \cup B$ and A^c, where the complement is relative to Ω. Decide whether $B \subseteq A$, giving a reason.

Part B — `W01-Q01-PB` (4 marks; 12 minutes) Define $f : \{-2, -1, 0, 1, 2\} \to \{0, 1, 4\}$ by $f(x) = x^2$. A student says, "f is one-to-one because every input has exactly one output." Determine whether f is injective and whether it is surjective for the stated codomain. (2 marks.)

For an integer n, let P be "n is even" and Q be "n^2 is even". Choose the correct relation: $P \Rightarrow Q$, $Q \Rightarrow P$, or $P \Leftrightarrow Q$. Justify the non-immediate direction by contradiction. (2 marks.)

Question 2 — quantifiers and induction

Part A — `W01-Q02-PA` (5 marks; 10 minutes) Negate the statement

$$\forall x \in \mathbb{R}, \quad x^2 + 1 > 0.$$

Then decide whether the original statement is true and justify your answer.

Part B — `W01-Q02-PB` (8 marks; 20 minutes) For $n \in \mathbb{N}$ with $n \geq 1$, prove by induction that

$$\sum_{k=1}^{n} (3k - 1) = \frac{n(3n + 1)}{2}.$$

Question 3 — logarithms and a declining benefit stream

Part A — `W01-Q03-PA` (6 marks; 15 minutes) Solve

$$\log(x - 1) + \log(x + 2) = \log 18$$

over the real numbers. State the domain restriction before manipulating the equation and check every candidate against it.

Part B — `W01-Q03-PB` (8 marks; 22 minutes) A programme produces a benefit of 100 units at $t = 0$. The benefit in each later year is $3/4$ of the preceding year's benefit.

1. State the index range and simplify $\prod_{i=1}^{4} r^i$ to a single power of r. (2 marks.)
2. Calculate the total benefit over an infinite horizon and state why the series converges.

3. Let N be the final included year, so the cumulative benefit covers $t = 0, 1, \ldots, N$. Find the smallest N for which cumulative benefit is at least 350 units. Use logarithms to solve the inequality. You do not need to calculate both adjacent numerical powers after finding the smallest admissible integer.

Question 4 — a logarithmic function

Part A — `W01-Q04-PA` (6 marks; 15 minutes) Let $g(x) = \log(9 - x^2)$. State the domain of g, then determine the set

$$\{x \in \mathbb{R} : g(x) \geq \log 5\}$$

in interval notation.

Question 5 — derivatives, an integral and marginal cost

Part A — `W01-Q05-PA` (6 marks; 14 minutes) For $x > 0$, let $f(x) = x^2 \log x$. Calculate $f'(x)$ and $f''(x)$. Also calculate

$$\int_1^e \log x \, \mathrm{d}x.$$

Part B — `W01-Q05-PB` (4 marks; 12 minutes) Suppose $C(q) = 60 + 4q + \frac{1}{2}q^2$ is a differentiable approximation to total cost in euros when $q \geq 0$ units are produced. A colleague says that $C'(6) = 10$ means "the seventh unit costs exactly 10 euros". Assess this claim. Explain the meaning of $C'(6)$ and compare it with the exact one-unit increase $C(7) - C(6)$.

Question 6 — Taylor approximation

Part A — `W01-Q06-PA` (8 marks; 20 minutes) Construct the first- and second-order Taylor approximations to $f(x) = \log x$ about $a = 1$. Use both to approximate $\log 1.2$.

Part B — `W01-Q06-PB` (3 marks; 9 minutes) For comparison, take $\log 1.2 \approx 0.1823$. Compare the absolute errors of your two approximations and explain what the comparison does, and does not, show about Taylor approximation away from its centre.

Question 7 — optimisation on an interval

Part A — `W01-Q07-PA` (8 marks; 20 minutes) A firm's profit is

$$\pi(q) = 18q - q^2 - 32,$$

where $q \in [0, 12]$. Find the global profit-maximising output and the maximum profit. Show the stationary-point and endpoint checks.

Part B — `W01-Q07-PB` (2 marks; 10 minutes) A student finds the stationary point and immediately calls it the global maximum. Explain what is missing from that argument on a closed interval and why the global conclusion is nevertheless valid in this problem.

Question 8 — probability as set algebra

Part A — `W01-Q08-PA` (8 marks; 15 minutes) For two events A and B, suppose

$$\mathbb{P}(A) = 0.58, \qquad \mathbb{P}(B) = 0.47, \qquad \mathbb{P}(A \cap B) = 0.31.$$

Calculate the probability of: (i) $A \cup B$; (ii) neither event; and (iii) exactly one of the two events.

Part B — `W01-Q08-PB` (2 marks; 8 minutes) A student reports $\mathbb{P}(A \cup B) = 0.58 + 0.47 = 1.05$. Identify the reasoning error and explain the correction in event terms.

Question 9 — a panel with a chair

Part A — `W01-Q09-PA` (7 marks; 18 minutes) Eight applicants are available for a three-person panel. The chair must be one of the three panel members. Two particular applicants, Ada and Bruno, may not both sit on the panel. How many distinct panel-and-chair outcomes are possible?

Question 10 — representation across proposal fields

Part A — `W01-Q10-PA` (10 marks; 25 minutes) A fund has 12 proposals: 5 in theory, 4 in applied economics and 3 in economic history. Four proposals are selected uniformly without replacement.

Calculate the probability that all three fields are represented. First count the selections by identifying which field supplies two proposals. Then give a second count using the complement and inclusion–exclusion, and reconcile the two results.

Self-test questions

Set aside 25 minutes and attempt all four parts without the reading guide, worked examples, hints or exercise solutions. No calculator is needed. Show the condition or definition that justifies each conclusion. The self-test has 20 marks and does not contribute to a checkpoint or mock-examination score.

Questions

`W01-ST01-PA` — functions, injectivity and surjectivity (`M1`; 5 marks; 6 minutes) Define

$$f : \{-1, 0, 1, 2\} \to \{0, 1, 4\}, \qquad f(x) = x^2.$$

Decide whether f is injective and whether it is surjective for the stated codomain. Explain why "every input has exactly one output" does not by itself establish injectivity.

`W01-ST02-PA` — a logarithmic equation (`M2`; 5 marks; 6 minutes) Throughout this part, log denotes the natural logarithm. Solve

$$\log(x - 2) + \log(x + 1) = \log 10$$

over the real numbers. State the domain before combining logarithms and check each algebraic candidate.

`W01-ST03-PA` — optimisation on a closed interval (`M3`; 5 marks; 7 minutes) Let

$$\pi(q) = 8q - q^2, \qquad q \in [0, 6].$$

Find the global maximiser and maximum value. Show the stationary-point, curvature and endpoint checks.

`W01-ST04-PA` — probability as event algebra (`S1`; 5 marks; 6 minutes) Suppose

$$\mathbb{P}(A) = 0.6, \qquad \mathbb{P}(B) = 0.5, \qquad \mathbb{P}(A \cap B) = 0.3.$$

Calculate $\mathbb{P}(A \cup B)$, the probability of neither event, and the probability of exactly one event.